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BO-HAN WU

Research Interests

Developing machine learning co-designed quantum systems to explore design spaces in quantum photonics and quantum information science. I am interested in leveraging machine learning to identify innovative solutions beyond human intuition, optimizing the performance and scalability of quantum technologies.

Professional Experience

2023 - 2025	Postdoctoral Researcher	<i>Research Laboratory of Electronics (RLE), Massachusetts Institute of Technology (MIT), Cambridge, MA, USA. Advisor: Prof. Dirk Englund</i>
2022	Visiting Student	<i>Electrical Engineering and Computer Science (EECS), University of Michigan, Ann Arbor, MI, USA. Advisor: Prof. Zheshen Zhang</i>
2014 - 2016	Research Assistant	<i>Institute of Atomic and Molecular Sciences (IAMS), Academia Sinica, Taipei, Taiwan Advisor: Dr. Ying-Cheng Chen</i>

Education

2017 - 2022	Ph.D., Physics,	<i>The University of Arizona (UArizona), Tucson, AZ, USA Advisor: Prof. Zheshen Zhang and Prof. Quntao Zhuang Thesis: Continuous-variable photonic quantum information processing</i>
2011 - 2014	M.S., Physics,	<i>National Tsing Hua University (NTHU), Hsinchu, Taiwan Advisor: Prof. Ite Yu Thesis: Studies of Transient Effects from Single-Λ to Double-Λ Systems under the All-Resonant Condition</i>
2007 - 2011	B.S., Electrophysics,	<i>National Chiao Tung University (NCTU), Hsinchu, Taiwan Advisor: Prof. Tien-Chang Lu Thesis: The Study of optimizing Photon Interaction in VCSEL with ITO material</i>

Grants Experience

2024	AFRL (FA8750-25-2-0500), Awarded , Co-write with Dr. A. Kumar and Prof. D. Englund
2023	DARPA-INSPIRED (HR001123S0052), Awarded , Co-write with Prof. M. Yu (USC), Dr. Z. Chen (USC), Prof. D. Englund (MIT) and Prof. Q. Zhuang (USC)
2023	HRI-MIT collaboration statement of work, Approved by Honda Research Institute (HRI) USA

Awards and Honors

- 2012 Shang-Keng Ma Scholarship: Outstanding Statistical Mechanics study (**Top 5%**)
- 2010 Movie Physics Competition: Creating videos that analyzed the physics in movies (**Top 5%**)
- 2007, 2008 Joint Calculus Examinations (I), (II): Calculus competition (**Top 2%**)

Invited Talks

- 2024 Cornell University (Department of Applied & Engineering Physics) special talk @ Zoom
- 2023 Seminar of Hon-Hai Research Institute @ Zoom
- 2023 NTHU (Department of Physics), Hsinchu, Taiwan
- 2023 National Yang Ming Chiao Tung University (Department of Photonics), Hsinchu, Taiwan
- 2023 Academia Sinica (Institute of Atomic and Molecular Sciences, Academia Sinica), Taipei, Taiwan
- 2022 MIT EECS seminar of Optical and Quantum Electronics @ Zoom

Journal Publications

- 2025 1. “Quantum Neuromorphic Sensor Network,” **B.-H. Wu**, S.-Y. Ma, S. K. Vadlamani, H. Choi, D. Englund, arXiv:2504.16119 (2025)
- 2025 2. “Fast Remote Quantum key distribution using single-photon absorption,” **B.-H. Wu**, H. Choi, D. Englund, in preparation
- 2024 3. “Formation of Quasi-Bound States in the Continuum in a Single Deformed Microcavity,” S. Liu, **B.-H. Wu**, J. Huang, Z. Zhang, Photon. Res. 12, 2651-2659
- 2024 4. “Distinguishing under-and over-coupled resonances without prior knowledge”, C. Cui, L. Zhang, **B.-H. Wu**, S. Liu, P.-K. Chen, L. Fan, Optica 11 (2), 176-177
- 2023 5. “Entanglement-assisted multi-aperture pulse-compression radar for angle resolving detection,” **B.-H. Wu**, S. Guha, Q. Zhuang, Quantum Sci. and Technol. 8 (3), 035016
- 2022 6. “Continuous-variable quantum repeaters based on bosonic error-correction and teleportation: architecture and applications”, **B.-H. Wu**, Z. Zhang, Q. Zhuang, Quantum Sci. and Technol. 7 (2), 025018
- 2020 7. “Quantum computing with multidimensional continuous-variable cluster states in a scalable photonic platform,” **B.-H. Wu**, R. N. Alexander, S. Liu, Z. Zhang, Phys. Rev. Research 2, 023138
- 2020 8. “Broadband coherent optical memory based on electromagnetically induced transparency,” Y.-C. Wei, **B.-H. Wu**, Y.-F. Hsiao, P.-J. Tsai, Y.-C. Chen, Phys. Rev. A 102 (6), 063720
- 2019 9. “Theoretical study of a memory-based optical converter with degenerate Zeeman states,” P.-J. Tsai, Y.-C. Wei, **B.-H. Wu**, S.-X. Lin, Y.-C. Chen, Phys. Rev. A 100 (6), 063843
- 2017 10. “Enhanced spectral profile in the study of Doppler-broadened Rydberg ensembles,” **B.-H. Wu**, Y.-W. Chuang, Y.-H. Chen, J.-C. Yu, M.-S. Chang, I. A. Yu, Sci. Rep. 7 (1), 9726
- 2016 11. “High conversion efficiency in resonant four-wave mixing processes,” C.-Y. Lee, **B.-H. Wu**, G. Wang, Y.-F. Chen, Y.-C. Chen, I. A. Yu, Opt. Express 24 (2), 1008-1016

Conference Publications

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| 2023 | 1. “Generation of Squeezed Light in Silicon Nitride Photonic Integrated Chips,” S. Liu, A. Hariri, B.-H. Wu , Y. Zhang, and Z. Zhang, <i>2024 Conference on Lasers and Electro-Optics (CLEO)</i> , Technical Digest Series, paper FF2H.6. |
| 2020 | 2. “Continuous-variable quantum repeaters based on bosonic error-correction and teleportation: architecture and applications,” B. -H. Wu , Z. Zhang and Q. Zhuang, <i>2022 CLEO</i> , San Jose, CA, pp. 1-2. |
| 2020 | 3. “Quantum-Computing Architecture based on Large-Scale Multi-Dimensional Continuous-Variable Cluster States in a Scalable Photonic Platform,” B.-H. Wu , R. N. Alexander, S. Liu, Z. Zhang, <i>2019 Conference on APS March meeting @ YouTube</i> . |
| 2017 | 4. “Electromagnetically induced transparency based light storage in short pulse regime,” B.-H. Wu , Y.-F. Hsiao, P.-J. Tsai, Y.-C. Chen, <i>2017 Annual Meeting of the Physical Society of Taiwan</i> . |
| 2014 | 5. “Studies of Conversion Efficiency and Energy Loss in Four-Wave Mixing Processes,” B.-H. Wu , C.-Y. Lee, G. Wang, Y.-F. Chen, Y.-C. Chen, I. A Yu, <i>2014 Annual Meeting of the Physical Society of Taiwan</i> . |

Journal Reviewer

Communication Physics, Phys. Rev. Research, Quantum Electronics, Applied Physics Letters, Entropy, Optical and Quantum Electronics.

Teaching Experience

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| 2023 | 6.2410 (Quantum Engineering System): MIT, Electrical Engineering and Computer Science |
| 2019 | MSE350 (Numerical Methods): UArizona, Materials Science and Engineering |
| 2018 | PHYS241 (Introduction to Electricity and Magnetism): UArizona, Physics |
| 2017-2018 | PHYS141 (Introductory Mechanics): UArizona, Physics |
| 2017 | PHYS181 (Introductory Laboratory I): UArizona, Physics |
| 2012 | Mathematical and Scientific Thinking Lecture (General Education): NTHU, Physics |
| 2012 | Daily Physics (General Education): NTHU, Physics |
| 2011 | Introduction to Electrodynamics (I), (II): NTHU, Physics |

Mentoring Experience

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| 2024 | Supervised MIT intern (UArizona PhD student, Isack Padilla) in the theoretical research in “Constructing quantum-augmented network” empowered by machine learning |
| 2023 | Supervised MIT undergraduate student (Hank Stennes) in Undergraduate Research Opportunity Program: building a prototypical digital twin of spin-spin cluster state generation code |
| 2022 | Supervised UArizona PhD students, John Mack and Abdulkarim Hariri (now EECS PhD student in the University of Michigan), in on-chip squeezing measurement and building up optical testing system. |
| 2021 | Supervised UArizona Master student, Zhaohui Yang (now CS PhD student in the University of California Santa Babra), in the research of quantum information theory |

Highlighted Research Projects

- 2023 **Quantum Neuromorphic Sensor network** (MIT): I proposed a quantum sensor-network structure, where each sensor nodes perform physics-based convolution on the RF signal to extract it feature information
- 2023 **Large-scale spin entanglement generation** (MIT): I developed digital twins using quantum Monte Carlo algorithm to generate high-fidelity nuclear electron-spin entanglement (>99%), simulating the nuclear spin entanglement generation with the proposed protocol (i.e. Barrett-Kok scheme and Humsphrey scheme)
- 2023 **High-rate quantum key distribution with single-photon** (MIT): I designed and simulated the quantum-key distribution (QKD) rate and the entanglement fidelity of the proposed atomic absorption-based quantum communication protocol
- 2021 **Quantum radar** (UArizona): I developed a radar scheme in the use of a transform-limited quantum microwave pulse field and demonstrated the appreciable quantum advantage (> 15 dB) by setting a finite pulse duration (0.1 second)
- 2020 **Quantum integrated photonic chip** (UArizona): I experimentally designed, fabricated and characterized the microstructure (e.g. ELS-7000, Versaline DSE III, MLA150, FC2500), and setting up free-space optical measurement system (Pound-Driver-Hall technique and homodyne measurement). In this experiment project, I achieved 0.83 dB level squeezing on silicon photonic platform (SiN). Moreover, I developed the theory in simulating the quantum dynamics in photonic chips to match the theory with experiments.
- 2016 **EIT based light storage** (IAMS): I studied the broadband optical memory based on the EIT protocol and provided a theoretical analysis of the requirements needed to achieve efficient and broadband EIT memory.
- 2014 **Rydberg EIT spectra under low atomic density** (NTHU): I investigated Rydberg-EIT spectra in room-temperature ^{87}Rb atoms, developing a theoretical model to describe how the contrast of EIT transparency changes with probe field intensity. Initially, the transparency contrast increases, peaks, and then gradually decreases. At optimal intensity, the contrast is 2-4 times higher than at the lowest intensity.
- 2012 **Efficient conversion in four-wave mixing process** (NTHU): I proposed an efficiency scheme of the resonant four-wave mixing (FWM) for the frequency up or down conversion. In this new scheme, two control fields are spatially varied such that a probe field at the input can be converted to a signal field at the output. The efficiency of probe-to-signal energy conversion can be 90% at medium's optical depth of about 100.
- 2009 **ITO material optimization in VCSEL** (NCTU): I used the Thin Film Calculator simulation tool to model the optical field eigenmodes in a Vertical Cavity Surface Emitting Laser (VCSEL). I optimized the VCSEL structure and ITO material thickness to improve lasing efficiency